

**Corporeal Capture: A Post-structuralist Reflection on Boundaries in *Crusader*
Kings III and *Civilization VI***

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Abstract

This article offers a philosophical reflection on the nature of boundary-drawing in software and video games. The concept of corporeal capture is offered to understand boundary as capable of localizing relational qualities (e.g., the quality of shyness is understood as internal to someone rather than co-produced in a distributed situation involving multiple social factors). We study two video games: *Crusader Kings III* and *Civilization VI*, to demonstrate how boundary-drawing in software can open up opportunities to inject assumptions and biases. The second half of the paper understands boundary-drawing as an inevitable part of doing creative work in software and games through its specifiability, controllability and scalability. We conclude by acknowledging software not purely as means of representation, but as mediums with their own peculiarities that need to be engaged with mindfully and critically.

Keywords: Boundary, *Crusader Kings III*, *Civilization VI*, Deleuze and Guattari, Post-structuralism

Introduction

The central topic of reflection in this article is boundaries. A boundary's basic function is to mark what is internal/external, intrinsic/extrinsic and inherent/extraneous. This includes the simplest example of spatial boundaries, from which one can make the inside/outside distinction. Beyond spatial boundaries, one can also claim a certain abstract quality as intrinsic to an entity, with this collection of intrinsic qualities defining what one might call *boundaried* being. For example, when one says "the die is fair," one interpretation is that the die has an internal quality of fairness, which is typically associated with the die's shape and weight distribution. This intrinsic relation could also be stated with the following pseudocode snippet:

```
class Die {
    bool fairness = true;
}
```

The above snippet creates an object class called *die*, with an internal attribute of type boolean (either true or false) named *fairness*. Die and fairness constitute a *has-a* relationship (class Die *has a* boolean value named fairness). This relationship is seen everywhere in software schema and video game systems: the player and enemies have health and magic bars; an item has a status buff; a user has a name and an ID etc. In fact, the object-oriented programming languages, commonly used in game development, make the creation and manipulation of boundaried entities their primary conceptual construct (Abadi & Cardelli, 2012). It is within this function of drawing internal/external distinctions that we wish to examine boundaries in software artifacts critically.

We situate this work within a larger philosophical investigation towards the nature of meaning making. In the context of software, there is a well-documented lineage from the

development of logical systems in analytic philosophy to their adoption as the basis for digital computers (Davis, 2018). The initial goal of formal logic was to develop a logical system to ground human meaning making by expressing meanings in logical expressions (Martinich & Sosa, 2001). Computational thinking thus enjoys a privileged status, often promoted as the foundation to other disciplines like science and mathematics (Sack, 2019, p.7). However, the logical approach to meaning making faced no small amount of criticism and many alternatives. Most notably, the later writing of Ludwig Wittgenstein explicitly criticizes the logical approach and cautions against grounding meaning in “calculi which have fixed rules” and “logic for a vacuum” (Wittgenstein, 2010). Separately, the structuralist tradition in philosophy also proposes a similar theory of meaning grounded within the relationality of its symbols (Dosse, 1997). Post-structuralism, developed in the second half of the 20th century, further embraces the grounding of meaning within relations and rejects the existence of a larger and stabilized system consisting of those relations (Dosse, 1998). The Wittgensteinian, structuralist and post-structuralist perspectives share a common rejection of the logical approach by defining meaning in terms of the relationality that situates symbols in a larger context that can’t be captured in a closed and formal system. Meaning making in this case is not only epistemological but also ontological — to explicate how we come to mean also implicates the nature of the entity that we come to represent.

This article takes a post-structuralist position, mostly inspired by works of French philosophers Gilles Deleuze and Félix Guattari (D&G) (Deleuze & Guattari, 2009, 1988), to reflect on the limit of a purely analytical and logical approach to represent human meaning. We wish to demonstrate that boundary-drawing is a point of failure for software and games as *logos* — as representations and stabilized reasoning. Specifically, boundaries can capture outside

interactions or objects into themselves through a process we term *corporeal capture*, introduced in the next section. We study two video games: *Crusader Kings III* (CK3) and Sid Meier's *Civilization VI* (Civ6) (ParadoxInteractive, 2020; FiraxisGames, 2016), and show how boundary-drawing can be used to inject cultural and ideological assumptions about the human body, ownership and the state.

Even though boundary-drawing can signal failure to represent real state of things, we argue that boundary-drawing is nevertheless necessary for creative works in software/game development. In the last main section, we connect the notion of boundary to the whole vs. parts distinction, and show how software/game boundaries afford specific ways for human meaning-making through its specifiability, controllability and scalability. Boundaries in software/games are thus different from those in human language due to their forms (artifacts of play and use) and materiality (the speed of computation). In conclusion, we argue that a responsible and joyful software/game practices should be conducted with this limitation of boundary in mind.

Corporeal Capture

As discussed, boundaries do not merely demarcate a space, but also capture a quality or an object originally outside of what is demarcated. The die example, where fairness is considered internal to the die's body, is the most basic case of capture. In the term corporeal capture we use the word "corporeal" because the process concerns bodies. We understand "body" in the most general sense, as a boundaried being, by having an explicit internal and external distinction. The body of a die is typically understood as spatially drawn, while the body of more abstract objects like "culture" is vague, but nevertheless boundaried. We inherit this term from D&G, who are

also quite loose with their definition of what a body can be. Representations can have their own bodies separate from the object that they are supposed to represent (Deleuze and Guattari, 1988, p.86) (one might think of signifiers, although D&G avoid the direct use of the term). The process of corporeal capture is illustrated in Figure 1.

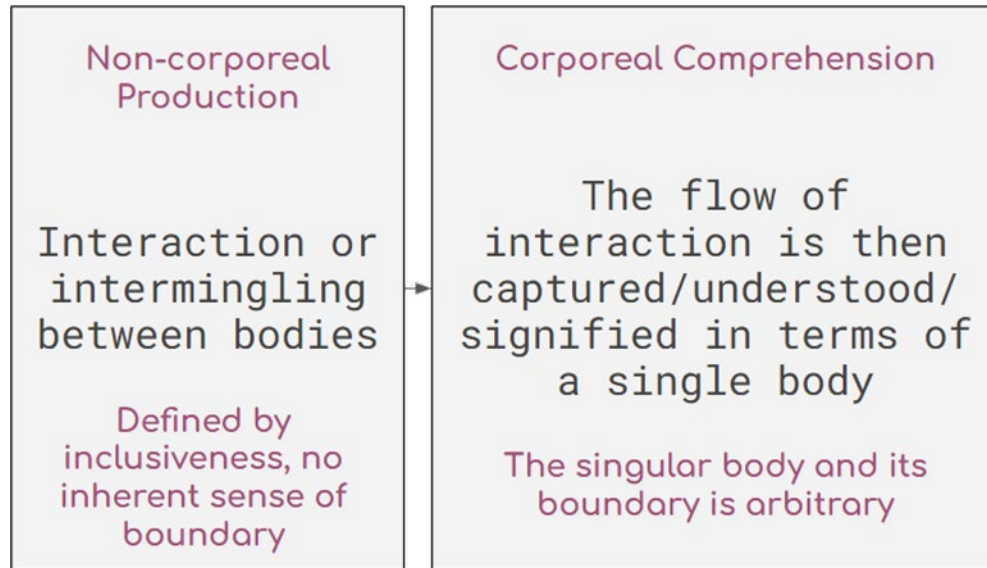


Figure 1. Overview of Corporeal Capture

Non-corporeal Production indicates a state where multiple entities or bodies interact with each other to (re)produce a certain quality or object. We use the word “production” here mostly from D&G’s usage in *Anti-Oedipus* (AO), “there is no need to distinguish here between producing and its product. We need merely note that the pure ‘thisness’ of the object produced is carried over into a new act of producing” (Deleuze and Guattari, 2009, p.7). When this article refers to a quality or an object, like fairness, it always refers to a process of production and reproduction. There is no inherent sense of boundary in (re)production. Objects and qualities are purely defined by their relations: the quality of fairness is created through the interaction between the die’s body and the multiple bodies within the environment — the die has to roll in the air, hitting floors and walls along the way and landing on a side. Even though fairness is

typically associated with the shape of the die, one cannot deny the multi-body environment required to (re)produce a die's fairness: the hand has to throw it with adequate force; the area has to be big enough for the die to bounce around etc.

Corporeal Comprehension denotes the state after corporeal capture takes place, where a particular boundaried entity captures the result of the non-corporeal production. "The die is fair" is comprehended *as if* the die body possesses the quality of fairness. The code snippet above as well as the statement "this is a fair die" explicitly makes this quality intrinsic to the die object. There are two notable aspects of corporeal capture. First, a *localization* underlies this process of capture: a body is seen to possess a quality that is in fact created by a more global and holistic interaction between multiple bodies. Hence, *corporeal capture denotes a capturing of a global product into a local body*. Second, the spatial and temporal specificity of the productive process is severed from the product. Let's define fairness as a mathematical distribution of outcomes over time (alternatively, one can also define the quality of fairness as produced through human perception of the die (Boschi et al., 2018)). The quality of fairness is (re)produced through rolling the die in an environment with specific physical, social and formal configurations. It is global (not localized to the die's body) as well as temporally and spatially specific, because the environment and the movement of the die are all involved within this production of fairness. Can one say a six-sided die is fair if it is stuck in a tube? But the notion of a "fair die" not only localizes this product, but it also separates the quality from parts of the context that produce it, as if fairness can be defined completely through a die's body.

Another example is when an accused is declared a "criminal." The object criminal and its qualities are produced and reproduced in an assemblage of bodies, whether it is the judges and the juries determining the result of criminal charges in the legal system, or the people producing

a colloquial impression of a criminal based on social conditioning and past experiences. The criminal quality is then *assigned to* or *captured by* the body of the accused as if the quality is intrinsic to their body. This is why D&G name this case as an example of *incorporeal transformation*: a transformation that doesn't change the nature of the bodies, but rather the dynamics of how they intermingle, "the transformation [is] applied to bodies but is itself incorporeal, internal to enunciation [... incorporeal transformations] are the [effect] of statements but are attributed to bodies" (Deleuze and Guattari, 1988, p.81-86). The concept of corporeal capture helps make this picture even clearer because 1) the non-corporeal production suggests that the very quality of the criminal is produced through a distribution of agents operating in diverse contexts: legal, social and psychological etc., and 2) corporeal comprehension suggests that such production is then localized and attributed to a body, as if the quality produced is intrinsic to it, severing it from the spatial and temporal specificity from which this quality comes about.

What corporeal capture reveals is that the code snippet shown at the beginning of this article, where the object *die* has a boolean of variable *fairness*, already carries some assumptions and hidden premises. The relational and structuralist view thus reveals the limit of the logical and analytical approach by destabilizing its boundary and what it reifies as intrinsic and inherent to an object. We will move on to examining the ludic forms of CK3 and Civ6 through this lens, to show corporeal capture's cultural and ideological dimensions.

Crusader Kings III: Human Bodies as Capture

I will turn to using the lens of corporeal capture to examine the grand strategy game *Crusader Kings 3* (CK3). CK3 puts the player in a meticulous simulation of the Middle Ages. The series is noted for its mixture of Sims-like roleplaying and classical strategy gameplay like developing settlements, growing armies and conquering territories. CK3 is described as Sims-like because instead of playing as a nation, the player is a specific person in the simulation, where they have to navigate interpersonal relationships, roleplaying with personalities and beliefs as well as dealing with family dynamics. Because of this mixture, the series is often praised for highlighting the “personal nature of rulership” (Devereaux, 2022; Franklin, 2014; Wardrip-Fruin, 2020, p.220), where the characters have their own interests and motivations instead of selflessly acting to serve an abstract state body.

In CK3, *traits* and *skills* are the primary ways the game system represents characters. Skills consist of six numerical values where each represents how good a character is at a category of activities. The categories are Diplomacy, Martial, Stewardship, Intrigue, Learning and Prowess. The main thing to note is that it is within the player’s interest to increase these skills as much as possible, as they influence almost all aspects of gameplay such as war, realm management and interpersonal dealings. Among many things, the *traits* of a character modify these values and are the main ways the player engages with roleplaying as their character. We will focus on discussing three types of traits: personality traits, congenital traits and physical traits. As their names suggest, personality traits are used to describe the personality of the character, such as Brave (bonus to Martial and Prowess and interpersonal opinions) and Lazy (small penalty to all skills). Congenital traits are traits that can be inherited by the children of the characters like Lisp and Pretty/Handsome, while physical traits describe certain physical aspects of the character like One-Eyed or Disfigured.

The trait system is an exemplary case of corporeal capture, as traits are contained within a character. Traits thus explicitly attribute certain qualities (laziness) to a boundaried entity (character). But personality traits are not independent of the situations in which they occur; a typical shy person may become gregarious when situated among people of common interests, or a person may become braver when acting for a cause they believe in. The word laziness itself already comes with many assumptions that need to be examined socially. Is a person's laziness a trait of their character or a response to a societal problem, similar to how the anti-work movements are in response to a hyper-capitalist society and the disillusionment of meritocracy (Lepore, 2021; Rosenblum, 2021)? This is not to say that there's nothing within a capturing body that causes those observations. What's important here is that *corporeal capture engages with an explicit hiding of the more distributed situation that brings those qualities about*; those qualities are thus perceived in a localized body rather than a more global and distributed whole. In the context of CK3, a shy person is shy everywhere, regardless of whom they are interacting with and what environment they are situated in.

The case becomes more troubling when looking at congenital traits and physical traits. As mentioned, congenial traits are traits that can pass down along the familial line while physical traits cannot. There are two issues with how CK3 implements these traits. First, many effects of these traits boil down to simple penalties to skills and interpersonal relations such as Dwarf subtracting 4 from Prowess and 20 from Attraction, or physical traits like One-Eyed reducing Attraction and Disfigured reducing Diplomacy. Disabilities in CK3 are thus often coded as intrinsically undesirable. Because the effects of these traits are internal to a character, the undesirability becomes independent of the global situation. Lispings will always mean the person takes a penalty with Diplomacy. Even though one might argue that the penalty is small and can

be mitigated by other traits that boost Diplomacy, what's troubling here is that the penalty to Diplomacy is coded as inherent to Lisper. Rosemarie Garland-Thomson proposed the term *misfit* to think of disability not as an essential bodily trait nor a purely social phenomenon, rather a misfitting between the body and the world in a particular space and time, thus “emphasizes context over essence, relation over isolation, mediation over origination” (Garland-Thomson, 2011). Similar to non- corporeal production, disability is always produced by a combination between a bodily feature (e.g., one- handedness) and a worldly feature (e.g., non-inclusive game design) that leads to a quality of misfitting (e.g., unable to play the game) that is highly specific in space and time.

Second, some congenital traits describe qualities that are rather subjective such as Ugly/Beautiful and Stupid/Intelligent. The same criticism levied at personality traits can be used here, as an ugly person is modeled as inherently ugly, independent of their societal context. But there's another corporeal capture happening. Since these traits are subject to inheritance, the qualities are also coded as inherent to genetics, independent of non-genetic interactions within a person's body. This understanding of genetics confuses phenotypic traits (observable traits influenced by genetics) and biological effects of genes (LarsPorsenna, 2022). This makes genetics predictable and controllable in CK3. Since the effects of these traits are also inherent and associated with skills, the system thus explicitly encourages selective breeding. It is not surprising, then, that a gameplay guide about congenital traits is named “How to Run a Successful Eugenics Program” (Zieley, 2021). Even though the title is clearly tongue-in-cheek, it highlights that neither the procedural rhetoric nor the presentation in CK3 provides critical distance from this narrow way of viewing genetics. The audience has to actively inject irony to foster a sense of remoteness from this point of view. A missing opportunity here is to model how

a cultural setting can change the effect of a certain bodily feature. As Oma Keeling points out in their critique of CK3, medieval France has a more institutional and cultural recognition of the blind (Keeling, 2020). One of the reasons is that France instituted blinding as punishment (Wheatley, 2010, p.29), resulting in more blind people compared to England, which shared many common medieval histories with France. But because these traits and their effects are captured within a character, the medieval culture and society that CK3 tries so arduously to simulate become contemporary and generic in regard to their attitudes towards disabled bodies.

By capturing qualities within a body, the ludic form of CK3 has the effect of localizing the player's understanding of those qualities to the capturing body. Again, non-corporeal production presupposes that the production of qualities such as fairness, criminality or smartness are global and without an inherent sense of boundary at first. The consequence of dissolving its boundary means that the mapping of its productive process is never finished. Corporeal comprehension, however, creates the finish line, the marker where the investigation ends. If criminality is inherent within a person's body, then it must be the mechanism within their body that causes it, such as their mental ailment or some sickness of the soul. If disability and its effects are inherent within a CK3 character, then there's no need to look further at outside causes as sources of production. The mapping of causal mechanisms becomes sufficient to stop at where the boundary is drawn, thus obscuring the exterior causal sources.

As mentioned in the prior section, an aspect of corporeal capture is to separate the product from its spatio-temporally specific process of production. The result is that the product no longer possesses a specificity within time and space (where? when? who? how?) and is conceived as a stable object. One additional consequence of this seeming stability is that it makes the tracking of its states possible and legible. For example, a character's personality traits rarely

change once they reach adulthood in CK3 except for education events, framed as the parent (the player) taking a stance on their children's behavior. The events provide choices to replace one of the child's traits with another. The corporeal capture of the character's body posits that personality traits such as shyness are retained by a character, not a co-produced quality between the character and its context. In addition, the boundary also regulates when and how the outside causal sources can interact with the body and change its content: educational events thus possess a knowable, traceable and legible influence. The last major section of this article will elaborate further on this regulation of causality as a necessary step in software and game practices.

Civilization VI: Ownership and Significant Causality

This section will focus on examining *Sid Meier's Civilization* (Civ) series through the lens of corporeal capture, focusing on the most recent entry *Civilization VI* (Civ6). Civ is a turn-based strategy game series where the player plays as one among a selection of leaders representing their respective civilizations (*Civ* for game titles, *civilization* for what the player plays as). Over 6000 years of human history, or 500 in-game turns typically, the player develops and expands their civilization and strives for being the first to achieve victory over other competing civilizations. The gameplay loop centers on exploring a randomly generated map, building and developing cities to extract resources. Those resources are then used for a variety of purposes such as more expansion, scientific research, military building and diplomacy. Different civilizations have advantages in different activities. For example, Mali gets bonuses for building cities and commercial districts around deserts. The game also offers different victory conditions, such as cultural, scientific and military victories that cater to different playstyles.

Civ is not a stranger to criticisms in how it portrays human societies and histories both in its ludic forms and its representations (Lundblade, 2020). As early as 2002, Christopher Douglas criticized Civ's division between civilizations and barbarians (non-playable factions that are hostile to civilizations by default), and its expansionist play loop; both of which justify a colonialist view of history (Douglas, 2002). Kacper Pobłocki further critiques Civ's linear representation of scientific progress and its emphasis on mastery over the game system as a despotic "illusion of ultimate agency [...] the game is constructed in such a way that we can only become all-powerful by becoming the US" (Pobłocki, 2003). Even further, Alexander Galloway precludes Civilization's ability to represent history as it "embodies the logic of informatic control itself [...] the diachronic details of lived life are replaced by the synchronic homogeneity of code pure and simple" (Galloway, 2004). We follow a similar line of cultural and philosophical criticism by examining Civ's ludic form using the concept of corporeal capture. This examination also clarifies two additional effects of boundary: *ownership* and *significant causality*. The next section will complicate these criticisms by considering that these identified meanings are not guaranteed to emerge during play.

Douglas' criticism of Civ's colonialist view, as well as Pobłocki's criticism of Civ's emphasis on the ultimate agency of the state, are closely related to another effect of corporeal capture, which we term *significant causality*: the boundary not only captures an object or a quality, but its body also lays claims to its productive process (condition of possibility, genealogy or the familial line). D&G term this "quasi-cause, the source and fountainhead and estuary of the apparent objective movement" (Deleuze and Guattari, 2009, p.194). It is a quasi-cause because the capturing body appears *as if* it causes the quality to happen. This is also implicit in almost all the examples given in the previous section. One can conceive of a die's

physical body possessing the quality of fairness, because its shape is the significant source of the production of fairness. CK3 captures shyness in a character's body, because one can conceive of shyness as originating from within their physical body. The notion of beauty and intelligence is captured in genetics because there's a common perception of genes significantly determining these qualities. These claims of significant causality are all "true consciousness of a false movement" that conceive the capturing bodies as quasi-causes (Deleuze and Guattari, 2009, p.194). The boundary of the capturing body thus marks a reasonable endpoint of investigation. It is not that outside influences are not acknowledged, but that they *do not matter as much*. This reinforces the logic of localization and the severance of spatio-temporal specificity — society, culture and history are thus obscured and deemed un-substantial. The claim of significant causality lies at the heart of the corporeal capture of the state body in Civ, which justifies its existence by reinforcing the agency and capacities of the state: both the state and the player appear to be a unified decision maker and the author of various activities such as diplomacy, internal development and military actions. The player playing as a state body, as is common in many strategy games, is simply another substantiation of the state body's capture. The deterministic aspects of the game, where actions and their effects have precise numeric value, also promote a hyper-rational, disembodied approach to civilization development (Voorhees, 2009), further reinforcing a colonist and "becoming-the-US" fantasy.

Another effect of corporeal capture, which has been implied throughout but not explicitly addressed, is the *establishment of ownership relation*. One comes to understand the body as not only the causal source of production, but also a body that *possesses* the captured object/quality. In Civ, the player owns and stockpiles resources, including those extracted from the environment such as oil and aluminum, and the production of the populus such as science, culture and gold

(resembling tributes and taxations). Both ownership relation and significant causality are used by D&G for their social critique in AO. They propose the term *socius* as a body that “[appropriates] for itself all surplus production, and arrogating to itself both the whole and the parts of the process, which now seem to emanate from it as a quasi cause” (Deleuze and Guattari, 2009, p.10). One example of the socius is what D&G call *the body of the despot*, which can be either the actual despot in a monarchy, or more abstract bodies such as god or the state. Despots are “the sole and transcendent public-property owner, the master of the surplus or the stock, the organizer of large-scale works (surplus labor), the source of public functions and bureaucracy” (Deleuze and Guattari, 1988, p.427). The despot, god and the state can legitimize their possession of lands and productions, because it is socially accepted that the despot is the significant factor that brings them into being: everything is under the purview of deities or kings/queens (Deleuze and Guattari, 2009, p.154). The people and their surroundings thus become directly subordinated to the despot in a hierarchy. One mechanism of subordination is the concept of infinite debt (“we owe everything to the despot”) and thus the establishment of tribute and taxation (Buchanan, 2008, p.105). In the following paragraphs, we will further examine how space, time, resources and productions are represented when they are sublimated under the ownership of the state body in Civ6.

Civ is often criticized for representing civilizations as timeless entities, as in, persistent and unchanging throughout history (Franklin, 2014). The player picks a predefined civilization with certain traits before the beginning of the game. Although the player develops technologies and cultures as the game progresses, their civilization’s bonuses, its names and appearances remain unchanged. This has consequences even beyond its ludic form, as youtuber KyleKallgreenBHH points out, many leaders for Germany throughout Civ existed long before

the nationality of a unified Germany existed (Kallgreen, 2016). In addition, as Muñoz highlights in Civ6, the terrain generation system has a bias to generate certain environmental features near the player based on their civilization's traits (more likely to be placed next to the desert if the player has desert bonus) (Bijsterveld Muñoz, 2022). Thus, the player's relation to "space and the environment has become more mediated by nationhood". Recalling corporeal capture's effect to sever a product from its specificity in time and space, Civilizations in Civ are no longer situated within a particular point in space and time. Their features are not molded and conditioned by space and time. Rather, they transcend both space and time and can divide them and lay claims to them.

This transcended status of a civilization is exemplified by how science and technology are implemented. In Civ6, each city outputs science as a numeric value, which is then summed into the total science output belonging to the civilization (the player). This science output is then used for unlocking technologies along a linear tech tree. This implementation of scientific achievement is both too fast and too slow at the same time. It is too fast insofar as to how it gets adopted within a civilization's spatial borders and units. Once a technology's research is completed, the bonuses (e.g., Replaceable Parts adding one additional food yield to farms) are immediately applied throughout the land. It is too slow insofar as the technology adoption stops completely and absolutely at the border. Science and technology, whose logic itself has no internal sense of national boundaries, become sublimated to the logic of the state body. The same goes for any resource directly owned by the player such as culture and money. The stockpiles are at once everywhere (within the state's boundaries) and nowhere (not a specific point within these boundaries).

The fact that competition is the central goal of Civ justifies such an extreme interpretation of ownership to the point of exclusion. Culture victory is also one of the win conditions, which requires the player to generate a certain amount of tourism. One source of generating tourism is to construct archaeological museums. The player needs to train archaeologists to unearth artifacts in the world so that it can be displayed. One of the “theming” bonuses is to display artifacts from the same era but different civilizations, thus directly encouraging players to “own” artifacts from other civilizations’ history. Shannon Martino criticizes Civ6’s depiction of archeology as “not only stereotypes of archaeology, but archaeology as a practice from the Victorian era” (Martino, 2021). In addition, civilizations in Civ6 can earn points towards recruiting Great People, which include Great Scientist, Great Writer or Great General. Once recruited, the Great People becomes a unit whose movement and effect are under the player’s control. Their benefits also exclusively belong to the player’s civilization. The game provides textual descriptions for this ownership relation. The “recruit” event description reads, “After deliberation, [Great People] chooses to bring their talents to [Player’s Civilization].” If the player has the game advisor enabled, the advisor will also say that the Great People is “inspired by our civilization.” This is another example where the ownership relation is reinforced by the claim of significant causality.

Questions of Meaning

All the above criticisms focus on the philosophical implications of Civ’s ludic form. But this raises the question: to what extent do these procedural rhetorics carry over to play itself? When the player interacts with the game, how do they process and absorb these biases, if at all? Diane Carr responds to Pobjocki’s and Douglas’ critiques of Civ by arguing that the critiques

based on rules and cultures cannot be naturally extrapolated to play (Carr, 2007). Zhou et al. also criticize the traditional proceduralist view by proposing that meaning creation in play is selective and requires maintenance socially (Zhou et al., 2022). The concept of corporeal capture highlights a particular ludic schema and its philosophical consequences. But this does not mean that the player is required to interpret their play this way. Rather, corporeal capture enables, in Stephanie Jennings' terms, "agentic modality [... videogames] as assemblages caught in churning currents of culture, politics, and power" (Jennings, 2022). There is thus inevitable sacrifice when assuming that meaning emerges purely from exclusively either the artifact or the context around it. The criticism of both CK3 and Civ6 so far makes those interpretations more likely and natural while others more obscure. The meaning making potential of video games and software cannot be localized in its software forms, but has to be understood relationally or infrastructurally: churning the current of culture and politics and guiding the process of connection-creation. The actual process of meaning production has to be understood as an assemblage, or a distributed situation that includes heterogeneous factors that go beyond the software artifact.

The questions of intervention on meaning making thus involve first deciding where in the assemblage this intervention is carried out. The very setting of play itself can be a site where beliefs and perceptions can be shifted and shaped to change the meaning making process. Many studies have investigated using Civ for historical education (Alexander, 2013; ERDEM and PAMUK, 2020; Peterson et al., 2013; Pagnotti and Russell III, 2012). The information retained through play is shown to be highly selective (ERDEM and PAMUK, 2020). This opens up the question on how educators can intervene in meaning making and steer the lessons taken from play away from what the artifact itself implies. In the modding community, Trevor Owens argues that Civ, far from force feeding its meaning to the player, provides the opportunity for

intellectual engagement of history (Owens, 2011). The very availability of mods and discussions of the game change the dynamics of how the ludic schemas in Civ are perceived.

If we are to intervene through the ludic forms, corporeal capture offers a particular direction in which to imagine alternatives. For example, the previous critique states that the existence of civilizations in Civ transcends both space and time. How, then, might one imagine an alternative, where the resonances and consistencies of the state are conditioned by space and time, so that both can threaten the state's integrity and presupposed unity? Our earlier critique mentions that stockpiled resources are everywhere within the state's boundary and reside at no particular place within that boundary. An alternative might be that the stockpiles have a place of storage (state capital), where accumulation needs to be supported through institutions and infrastructures, so that tributes and taxes can be collected and transported efficiently. Borders and boundaries need to be constantly maintained and (re)defined — such as border controls for spatial movement and federal intellectual property for scientific discoveries. Thus, the management of speed becomes crucial for the maintenance of the state body: the increase of speed within its boundaries and the regulation of speed between its inside and outside. For the next section, we will generalize these suggestions into more abstract categories, in order to discuss how corporeal capture can inform game/software design and practices.

The Whole and its Parts

This last major section explores the design and other practical implications of the discussion around boundary and corporeal capture. The theoretical starting point is the whole vs. part distinction. It is not difficult to see how boundaries are relevant to the relationship between

parts and wholes. The whole vs. part analysis is often used to promote scientific and system thinking, as well as for software development and game design (Sellers, 2017). As commonly understood, the whole is formed by drawing a boundary around its parts, in which the parts can be understood as internal to and make up the whole. The concept of emergence is one key concept in this two-level analysis, as “[t]he whole has its own properties and behaviors belonging to the group but not to any single part within it”, just as different letters form a word, which embeds a very different and emergent meaning than its individual parts (Sellers, 2017, p.50). We wish to bring further complication to this understanding of the whole vs. part distinction. By pointing out the arbitrariness of boundaries and the effects of corporeal capture, we argue that *the whole is always created and maintained by a third party*. The previous two sections critique this existence of the third party as creating openings to inject assumptions and biases. But this section envisions this third party as a mindful practitioner engaging in necessary boundary-drawing, being aware of their positionalities and the practical, formal and material constraints of software practices.

As stated, the relationship between the whole and its parts is not simply one of encapsulation. The whole is upheld entirely separately from the parts. This view is shared also by D&G in AO, “We believe in totalities that are peripheral [...] it is a whole of these particular parts but does not unify them; rather it is added to them as a new part fabricated separately” (Deleuze and Guattari, 2009, p.42). Latour et al.’s provocatively named article “The Whole is always smaller than its Parts” also claims that “there is more complexity in the elements than in the aggregates” (Latour et al., 2012). The whole does not completely encompass its parts, rather, defining a whole is always a selective process on the parts that it wishes to encompass.

Emergence is always emergent to an outside. The introduction of a whole always implicates another party.

To repeat, non-corporeal production has no intrinsic sense of boundaries. This means that production, when viewed on the level of parts, fundamentally challenges the naturalness of the boundaries of the wholes, if not completely renders them unnecessary. For the “die is fair” example, the body of the die can be broken down to parts, such as vertices and edges in its 3D model. Additional information about its weight distribution can also be provided. The die’s body is thus split into different parts, not unified within a boundaried entity like the class *die*¹. Fairness will thus be produced through rolling in a physics simulation, returning its production back to the global interaction and intermingling between bodies. This also highlights how production is always *selective* within a given boundary: certain vertices of the die’s model and certain parts of the environment are relevant in a particular roll but not in other rolls. This selective nature challenges the boundary’s naturalness and necessity. The fairness must be defined by how the die is rolled, in what environment, over how much time and how many rolls.

For Civ6, one can imagine science and technology escaping the state logic by making the player lose some control of technological development, and adding different causal reasons like geographical proximity and information-sharing infrastructures like conferences or the internet. Thus, scientific development is driven by a spatio-temporal specificity: where the state is positioned, what enables certain scientific knowledge to spread in what way, and at what

¹One can still store this 3D model under the abstract object *die*, but at that point the boundary takes a much less important role in the production and simulation of fairness.

particular time. The parts thus follow their own logic to make connections regardless of where the boundary is drawn. The parts do not need the boundary to flow and produce. Instead, the boundary has to impose its own logic on the parts that are already flowing and producing (boundary of the die's body, federal intellectual property, border control). In this view, the whole is not an inevitability in aggregating the parts, just as Latour et al. put it, “you don't go from the particular to the general, but from particular to more particulars” (Latour et al., 2012).

On the other hand, modeling productive processes on the level of wholes always presupposes boundaries of the wholes. This is not in itself problematic because 1) the parts/wholes distinction is completely relative — one has to commit to a certain level of abstraction eventually, and 2) it is sometimes unintuitive, infeasible or undesirable to expose the parts in a representational schema. We will spend the rest of this section discussing how the inevitable act of boundary-drawing (whole-making) cannot be separated from the practical issues and constraints of software and game development. We will continue to follow the examples of CK3 and Civ6, and break down our points to three different factors: specifiability, controllability and scalability.

Interactions on the level of wholes are *specifiable* precisely because the designers can author them by regulating when, where and how the parts can interact. Imagine how one would model a die roll, for example, where the character determines the likely outcome of the roll: die + character → distribution of outcomes (similar to *Super Mario Party* (NDcube, 2018)).

Compared to how one might develop rolling strategies kinesthetically when using the mouse to throw dice in a simulation (throwing at what angle, hitting what corners, at what height), the former specifies the dynamics of production, making them legible and interpretable, while the latter is a simulation of the productive process whose dynamics is always changing and shifting:

sometimes the power of the throw dominates other factors, sometimes the height is not enough for a fair throw. The specifiability of the whole-mediated interactions enables the designer to regulate the causal flow of production and each element's causal power in production precisely because each element's boundary is presupposed, and the production is mediated through them.

The parental event mentioned previously in CK3 is also an example of this specificability, where the events are the explicit procedures to change a character's personality trait, making the causal dynamics of personality change specific, legible and interpretable. The earlier critique of CK3 mentions the trait of Blindness, where the negative effect of blindness is considered internal to the trait itself, obscuring more global production of disability involving cultural, material and social prejudices. A feasible adjustment would be to consider changing the negative effects of the Blindness trait based on which culture the character is situated in, such as the character losing more Diplomacy in French societies because the blind was often depicted in humorous situations in French literature (Wheatley, 2010, 90). This proposal still presupposes the boundaries such as the French and Anglo-Saxon cultures, and the separation between the character and the culture, thus losing the subtlety of their continuities. What's important in carrying out this exercise is showing that destabilizing the boundaries opens up possible venues of investigation and alternative ways of forming boundaries. Boundary drawing is thus also the *selection of causal chains* — a decision to explicate certain ones while closing off others.

The whole-mediated interactions become *controllable* because they always cut off the causal chain and localize the causal source. Previously, the interactions on the level of parts were characterized as possessing “their own logic to make connections regardless of where the boundary is drawn.” But this is not always desirable. For game authoring and development, it is critical to control the complexity of the simulation, where significant causal sources can be

located easily for iteration. This is closely related to the traceability and legibility of the productive process, where system dynamics and variability have to be carefully balanced with interpretability. Significant causality becomes a practical necessity in engineering.

Controllability also touches on the fundamental features of play — that is, involving a player and the played object. Latour et al. talk about the notion of the “dispatcher”

[The dispatcher] implies that the structure emerging out of interactions between atomistic agents should, in the end, emulate what this (absent) dispatcher was supposed to do: namely, to provide orders and rules of behaviour to the agents, even though, everyone agrees, there is no such order-giving entity. (Latour et al., 2012)

Latour et al. assert that the construction of the whole (emergent structure) over the parts (interactions between atomistic agents) instills an invisible dispatcher, or a unified decision maker, that acts in a top-down manner to ensure the emergence of intended behavior. This resembles the software/game form as an interactive medium, which necessitates the player-game or the user-software relation. Under this light, the player/user becomes the concrete embodiment of this unified decision maker and the localized source of significant causality. This affirmation of the player/user as a significant causal source can manifest in many ways and, specifically, mechanically and narratively in games. We have demonstrated how Civ6 presents the player as a dispatcher that exists everywhere (within its boundary) and nowhere (not a particular point in that boundary) to capture production and to order development of the state. Jennings critiques the monomythical “world savior” trope in videogame writing that reinforces an authoritarian form of player agency (Jennings, 2022). *The Elder Scrolls V: Skyrim* (BethesdaGameStudios, 2011)

centers the significant causal power of the player as its central narrative fantasy, allowing the player character to become the top member of most factions and a central part of most significant story events. The formal structure of software/game (player-played, user-used) makes controllability of the whole (the player/user as a unified decision maker) a necessary part of the creative process that implicates both the player and the designer.

The whole-mediated interactions become *scalable* because the stabilized boundary can obscure how the internal parts interact, allowing the system to function without exposing the parts. Sometimes the parts are simply implied, such as how a “person” object allows the player to imagine an inner dynamic without the system explicitly modeling those dynamics. But it is also relevant in software design itself. The code snippet at the beginning, where the object *die* has the attribute *fairness*, follows the paradigm of object-oriented programming (OOP). Alan Kay, a pioneer of OOP, wrote that the most important contribution of OOP is its “messaging, local retention and protection and hiding of state-process, and extreme late-binding” (Kay, 2003). For example, when one wants the *die* object to execute certain actions (e.g., *die 1.roll()*). The *roll* action would itself be a message sent to an instance of the *die* object from some external source (the user or another part of the system). What is important here is that because *die* has all the relevant data for a roll encapsulated within itself (protection and hiding), the *roll* action can simply be a token sent to the object as a message. The token message itself does not need to know about the structure and state of the *die* object, simply that the message will assume a “dispatcher” within the object *die* to set relevant internal interactions in motion (late-binding). This paradigm not only reduces the number of assumptions that programmers have to make to utilize the *die* object, but also allows a more efficient and scalable communication between objects. Boundary-drawing thus also has a material dimension that concerns itself with a

management of the cognitive load of the developer and computational load in the communication between objects.

Conclusion

In critiquing both CK3 and Civ6, this article has mostly focused on the failure of their software schema to represent the real state of things — their failure as logos. The concept of corporeal capture locates the origin of this failure at the point of boundary-drawing. The second half of the article introduces a subtler point, where we discuss how the representations in Civ6 can be subverted by the context of play, and how modeling based on the level of wholes is necessitated by its specifiability, controllability and scalability. Under this light, boundary drawing in software is functionally reductive: it is functional in so far as it is reductive.

One might notice that the critique of boundaries is also applicable to human discourse in general. But boundaries in software are of a completely different nature because of its formal and material specificities. This fundamental difference makes boundaries in software mutable and capable in certain ways while not others. What we seek is to recover a certain pleasure within practice, which comes with the recognition of one's responsibility in boundary drawing (Haraway, 2013), and to dispel a certain shame towards practice by interpreting boundary drawing as points of failure. This is not to say that computational modeling cannot take inspiration from what it's supposed to resemble. But what is important is to also recognize the irreducibility of software itself, as specified by its formal and material qualities. To lose sight of this is to fail at both ends: failing at recognizing the complexity of the objects of representation at one end, and the potential and the peculiarities of the software at the other.

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